Artificial Intelligence and Intelligent Medicine

Homework 1: Time-Series Data Analysis

Exploring EHR Data with PhysioNet 2012

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P1. Preprocessing

(a) Data structure

`set-a/132539.txt` has shape $(273, 3)$. In row 7, Time is 7 minutes after ICU admission, Parameter is heart rate, and Value is 73 bpm.

Table 1: Raw data for patient 132539

Index	Time	Parameter	Value
0	00:00	RecordID	132539
1	00:00	Age	54
2	00:00	Gender	0
3	00:00	Height	-1
4	00:00	ICUType	4
5	00:00	Weight	-1
6	00:07	GCS	15
7	00:07	HR	73
8	00:07	NIDiasABP	65
9	00:07	NIMAP	92.33

We used method (ii), scanning all 4,000 Set A files. The general descriptors are `RecordID, Age, Gender, Height, ICUType, Weight`. The 37 time series variables are `ALP, ALT, AST, Albumin, BUN, Bilirubin, Cholesterol, Creatinine, DiasABP, FiO2, GCS, Glucose, HCO3, HCT, HR, K, Lactate, MAP, MechVent, Mg, NIDiasABP, NIMAP, NISysABP, Na, PaCO2, PaO2, Platelets, RespRate, SaO2, SysABP, Temp, TroponinI, TroponinT, Urine, WBC, Weight, pH`.

Weight is recorded at admission and later, so it belongs to both groups. RecordID is excluded from model features.

(b) Relative time conversion

Source: `set-a/132539.txt`.

Output: `132539_hours.txt`.

We convert Time using $HH + MM/60$. All 273 rows, their order, Parameter, and Value are preserved.

Table 2: Examples of time conversion, with hours rounded for display

Time	Hours	Parameter	Value
00:00	0.000000	RecordID	132539
00:00	0.000000	Age	54
00:07	0.116667	GCS	15
00:07	0.116667	HR	73
00:37	0.616667	HR	77

(c) Time series structure

Running `P12_preprocess.py` on all patients produced (93,698, 45) for Set A and (93,685, 45) for Set B, with 4,000 patients each. Patient 132539 changes from $(273, 3)$ to $(24, 45)$. Table 3 shows its first six rows.

Table 3: Processed patient 132539. Entries denotes `seg_entry_cnt`

Hour	HR	GCS	Age	Weight	Target	Entries
2	70	15	54	-1	0	4
4	71	15	54	-1	0	3
6	73.5	`NaN`	54	-1	0	2
8	64	15	54	-1	0	1
10	65	`NaN`	54	-1	0	2
12	59.5	15	54	-1	0	2

Missing values are `NaN`. Empty bins have no rows. Measurements within each bin are averaged. The first nonmissing bin value of each static descriptor is copied to every row, overwriting later Weight values. Height values of -1 become `NaN`.

`target` is in-hospital death (0: survived, 1: died). `seg_entry_cnt` counts distinct raw timestamps per bin. Bins are 2 hours wide, cover $(t-2, t]$, and are labelled by t. Hour 0 is included in bin 2.

(d) Window selection

Keeping $12 \leq$ `timestamp` ≤ 36 reduces patient 132539 from 24 to 13 rows. We predict at hour 36 to collect enough data while leaving time to respond. Starting at hour 12 aims to reduce effects of initial treatment. The retained bins cover raw times $(10, 36]$.

Table 4: Patient 132539 before and after window selection

Before: 24 rows

Hour	HR	GCS
2	70	15
4	71	15
6	73.5	NaN
8	64	15
10	65	NaN
12	59.5	15
14	59	NaN
16	66	15
18	63	NaN
20	68.5	15
22	67.5	NaN
24	68.5	15
26	67	NaN
28	80	15
30	80	NaN
32	78	14
34	85	NaN
36	83	15
38	84	NaN
40	73	15
42	75	NaN
44	73	15
46	84	NaN
48	86	15

After: 13 rows

Hour	HR	GCS
12	59.5	15
14	59	NaN
16	66	15
18	63	NaN
20	68.5	15
22	67.5	NaN
24	68.5	15
26	67	NaN
28	80	15
30	80	NaN
32	78	14
34	85	NaN
36	83	15

(e) Missing value imputation

We replace Weight and Gender values of -1 with NaN, then forward fill and backward fill within each patient using observations available by hour 36. Remaining gaps use Set A medians for continuous features and modes for Gender and ICUType. Medians reduce outlier effects. MechVent is assumed 0 if absent throughout the window. Set B uses the same fallback values. Patient 132539 changes from 369 NaNs to none.

Table 5: Selected rows for patient 132539 before (b) and after (a) imputation

Hour	GCS (b)	GCS (a)	BUN (b)	BUN (a)	Weight (b)	Weight (a)
12	15	15	13	13	-1	78.4
14	NaN	15	NaN	13	-1	78.4
32	14	14	NaN	13	-1	78.4
34	NaN	14	8	8	-1	78.4
36	15	15	NaN	8	-1	78.4

(f) Zero padding

We exclude 8 patients per set with no window data and pad the remaining patients to 13 time points. Missing bins receive zero feature values and `seg_entry_cnt`=0. Metadata is preserved. Patient 132772 changes from 7 to 13 rows. "Absent" denotes a missing row. Both feature arrays have shape $(3992, 13, 41)$.

Table 6: Patient 132772 before (b) and after (a) padding

Hour	HR (b)	HR (a)	Age (b)	Age (a)	Entries (a)
12	124.5	124.5	68	68	1
14	Absent	0	Absent	0	0
16	124.5	124.5	68	68	1
18	Absent	0	Absent	0	0
20	Absent	0	Absent	0	0
22	Absent	0	Absent	0	0
24	Absent	0	Absent	0	0
26	124.5	124.5	68	68	1
28	Absent	0	Absent	0	0
30	124.5	124.5	68	68	3
32	100	100	68	68	16
34	93	93	68	68	10
36	88	88	68	68	5

(g) Labels and class distribution

Each retained patient has one label, ordered to match X. Both label arrays have length 3,992.

Table 7: Class distribution among retained patients

Class	Set A count	Set A proportion	Set B count	Set B proportion
0: survived	3438	0.8612	3424	0.8577
1: died	554	0.1388	568	0.1423

P2. Baseline training

Data preprocessing

We use P1 windowed data without imputation or padding. For 36 dynamic variables, we calculate last observed value, mean, minimum, maximum, and observed-bin count. Adding 5 static descriptors gives 185 features. IDs, targets, timestamps, and `seg_entry_cnt` are excluded.

Method

LightGBM handles missing values and tabular data. We use binary log loss, a small learning rate, limited leaves, subsampling, and an L2 penalty to help control overfitting.

Table 8: LightGBM settings

Parameter	Value
`n_estimators`	500
`learning_rate`	0.02
`num_leaves`	15
`min_child_samples`	20
`subsample` / `subsample_freq`	0.8 / 1
`colsample_bytree`	0.5
`reg_lambda`	1.0
`deterministic` / `force_col_wise`	True / True

We compare inputs using five stratified folds on Set A, fitting P1 fallback values within each training fold. Summary features scored higher in out-of-fold (OOF) evaluation, so we used them.

Table 9: Set A OOF comparison using seed 42

Input	AUROC	AUPRC
Flattened P1 inputs (13×41)	0.8317	0.4522
Summary features (185 features)	0.8378	0.4782

Results

Training on Set A with seed 42 gives **Set B AUROC 0.8545 and AUPRC 0.5122**, achieving the Strong tier. AUPRC uses average precision [3]. Set B is used for evaluation only.

Table 10: Set B results across all tested seeds

Seed	AUROC	AUPRC
42 (final)	0.8545	0.5122
7	0.8566	0.5202
123	0.8547	0.5097
256	0.8552	0.5152
1001	0.8533	0.5026
Mean	0.8549	0.5120
SD	0.0012	0.0065

Reproduction

All seeds used in P2 and P3 are **42, 7, 123, 256, and 1001**. The final P2 seed is **42**. Environment: Python 3.12.13, NumPy 2.5.4, pandas 3.1.1, scikit-learn 1.9.1, LightGBM 4.7.0, and Matplotlib 3.11.2, using CPU.

Place the notebook, `P12_preprocess.py`, outcomes files, and data folders or archives in `/content`. Run the notebook's installation cell once and restart the runtime. Run the remaining notebook cells in order.

P3. Feature resolution

Question

How much numerical detail does mortality prediction need?

Approach

We keep P2's LightGBM settings and compare original inputs with 2 to 8 quantile levels on Set A. Quantiles avoid manually choosing cutoffs for each feature. Gender, ICUType, counts, and missing values stay unchanged. Boundaries use training folds only. We use five stratified folds and five seeds, with matching folds across settings within each seed. Results show mean OOF scores and standard deviations. Differences use each seed's original inputs as the baseline.

Experiment

Individual seed scores are in the appendix.

Table 11: Set A OOF results: mean ± SD across five seeds

Levels	AUROC	AUPRC	Mean ΔAUPRC
Original	0.8364 ± 0.0027	0.4698 ± 0.0107	$+0.0000$
2	0.8048 ± 0.0028	0.4035 ± 0.0074	-0.0664
3	0.8207 ± 0.0017	0.4301 ± 0.0097	-0.0397
4	0.8254 ± 0.0023	0.4509 ± 0.0067	-0.0189
5	0.8291 ± 0.0012	0.4478 ± 0.0050	-0.0221
6	0.8334 ± 0.0024	0.4585 ± 0.0061	-0.0113
7	0.8372 ± 0.0026	0.4648 ± 0.0093	-0.0050
8	0.8327 ± 0.0034	0.4658 ± 0.0081	-0.0040

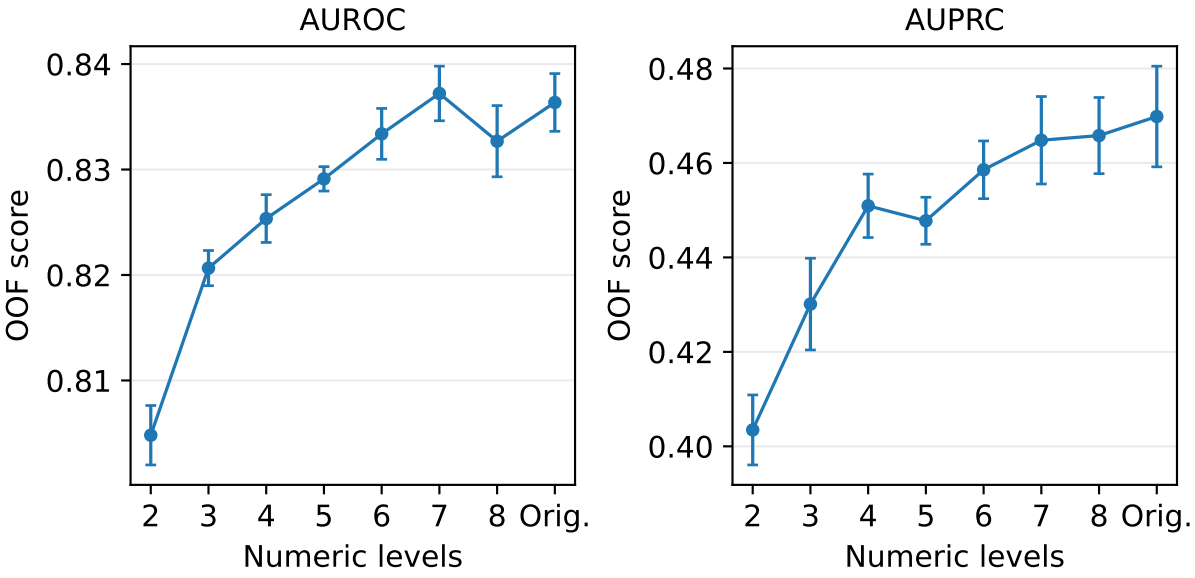


Figure 1: Numerical feature resolution and Set A OOF performance. Error bars show standard deviations across seeds, not confidence intervals. Orig. denotes the original inputs.

Two levels reduced mean AUPRC from 0.4698 to 0.4035. Seven and eight levels reached 0.4648 and 0.4658.

Conclusion

Eight levels performed similarly to the original inputs in this Set A experiment. This may be because LightGBM already bins numerical values [2].

References

[1] PhysioNet. Predicting Mortality of ICU Patients: The PhysioNet/Computing in Cardiology Challenge 2012, version 1.0.0.

[2] LightGBM documentation. Features and Missing value handling.

[3] Scikit-learn documentation. Average precision score.

Appendix. P3 scores by seed

Set A OOF scores for each seed and setting.

Table 12: Individual P3 seed results

Seed	Input	AUROC	AUPRC
42	Original	0.8378	0.4782
42	2 levels	0.8089	0.4153
42	3 levels	0.8211	0.4348
42	4 levels	0.8270	0.4540
42	5 levels	0.8275	0.4404
42	6 levels	0.8345	0.4619
42	7 levels	0.8395	0.4757
42	8 levels	0.8368	0.4723
7	Original	0.8323	0.4513
7	2 levels	0.8014	0.3997
7	3 levels	0.8181	0.4162
7	4 levels	0.8216	0.4420
7	5 levels	0.8286	0.4477
7	6 levels	0.8294	0.4488
7	7 levels	0.8338	0.4523
7	8 levels	0.8276	0.4517
123	Original	0.8356	0.4719
123	2 levels	0.8037	0.3979
123	3 levels	0.8217	0.4275
123	4 levels	0.8264	0.4468
123	5 levels	0.8305	0.4487
123	6 levels	0.8330	0.4613
123	7 levels	0.8363	0.4600
123	8 levels	0.8318	0.4679
256	Original	0.8397	0.4722
256	2 levels	0.8040	0.4061
256	3 levels	0.8200	0.4296
256	4 levels	0.8248	0.4523
256	5 levels	0.8298	0.4544
256	6 levels	0.8345	0.4567
256	7 levels	0.8364	0.4644
256	8 levels	0.8335	0.4682
1001	Original	0.8363	0.4755
1001	2 levels	0.8060	0.3983
1001	3 levels	0.8223	0.4425
1001	4 levels	0.8269	0.4595
1001	5 levels	0.8292	0.4476
1001	6 levels	0.8355	0.4641
1001	7 levels	0.8401	0.4716
1001	8 levels	0.8338	0.4688